STEP 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import classification_report
```

STEP 2: Load Dataset

```
df = pd.read_excel("/content/Dataset_Final_Project from PPDA.xlsx")
```

```
# View first rows
print(df.head())
```

```
        Contract_ID                          Procuring_Entity  \
0  TENDER-2024-1000             Uganda Revenue Authority (URA)
1  TENDER-2024-1001          Mulago National Referral Hospital
2  TENDER-2024-1002                        Makerere University
3  TENDER-2024-1003           Ministry of Education and Sports
4  TENDER-2024-1004  Ministry of Energy and Mineral Development

                   Category    Item_Description  Procurement_Method  \
0      Consultancy Services      Legal Services  Direct Procurement
1  Non-Consultancy Services   Catering Services  Direct Procurement
2  Non-Consultancy Services   Security Services  Restricted Bidding
3      Consultancy Services  Engineering Design  Restricted Bidding
4                  Supplies    Office Furniture  Direct Procurement

   Contract_Value_UGX  Number_of_Bids    Region     Status
0           443734972               1   Eastern  Completed
1            96090292               1  Northern  Completed
2           153171095              12   Western    Ongoing
3           252968651              10   Central  Completed
4            67250665               1  Northern  Completed
```

```
# Check structure
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1100 entries, 0 to 1099
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Contract_ID         1100 non-null   object
 1   Procuring_Entity    1100 non-null   object
 2   Category            1100 non-null   object
 3   Item_Description    1100 non-null   object
 4   Procurement_Method  1100 non-null   object
 5   Contract_Value_UGX  1100 non-null   int64
 6   Number_of_Bids      1100 non-null   int64
 7   Region              1100 non-null   object
 8   Status              1100 non-null   object
dtypes: int64(2), object(7)
memory usage: 77.5+ KB
None
```

STEP 3: Data Description

```
# Summary statistics
print(df.describe())
```

```
       Contract_Value_UGX  Number_of_Bids
count        1.100000e+03     1100.000000
```

```
mean         7.184356e+08        6.212727
std          1.241328e+09        4.404091
min          6.472542e+06        1.000000
25%          8.735007e+07        1.000000
50%          1.628312e+08        6.000000
75%          4.370229e+08       10.000000
max          4.964156e+09       14.000000
```

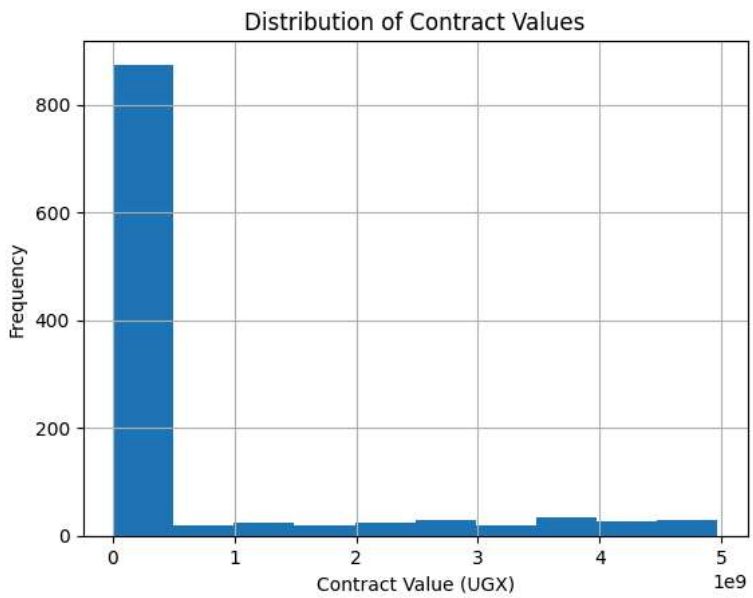
```
# Check unique values
print(df['Status'].value_counts())
print(df['Procurement_Method'].value_counts())
```

```
Status
Completed     761
Ongoing       277
Terminated     62
Name: count, dtype: int64
Procurement_Method
Request for Quotations        243
Direct Procurement            222
Open International Bidding    222
Restricted Bidding            212
Open Domestic Bidding         201
Name: count, dtype: int64
```

STEP 4: Data Visualization

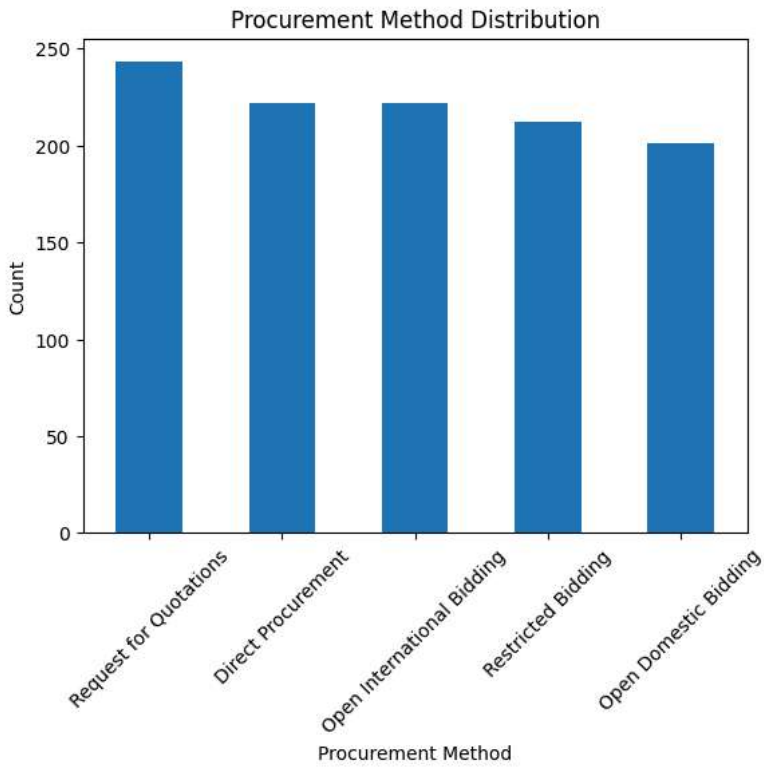
4.1 Univariate – Contract Value Distribution

```
plt.figure()
df['Contract_Value_UGX'].hist()
plt.xlabel("Contract Value (UGX)")
plt.ylabel("Frequency")
plt.title("Distribution of Contract Values")
plt.show()
```



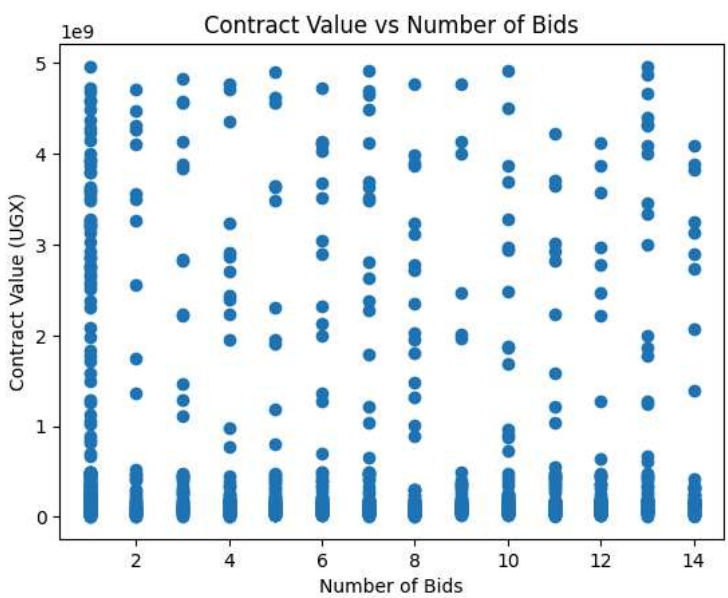
4.2 Univariate – Procurement Method

```
plt.figure()
df['Procurement_Method'].value_counts().plot(kind='bar')
plt.xlabel("Procurement Method")
plt.ylabel("Count")
plt.title("Procurement Method Distribution")
plt.xticks(rotation=45)
plt.show()
```



4.3 Multivariate – Value vs Bids

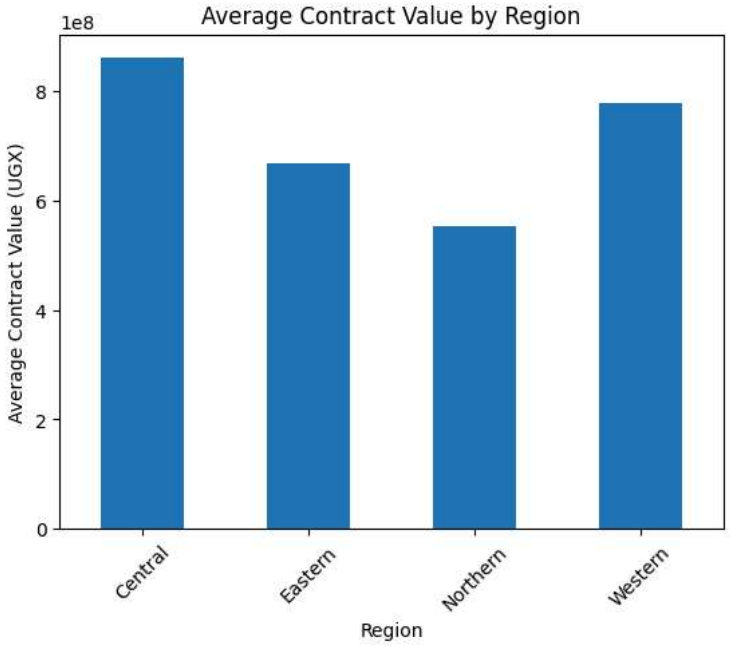
```
plt.figure()
plt.scatter(df['Number_of_Bids'], df['Contract_Value_UGX'])
plt.xlabel("Number of Bids")
plt.ylabel("Contract Value (UGX)")
plt.title("Contract Value vs Number of Bids")
plt.show()
```



4.4 Multivariate – Region vs Value

```
plt.figure()
df.groupby('Region')['Contract_Value_UGX'].mean().plot(kind='bar')
plt.xlabel("Region")
plt.ylabel("Average Contract Value (UGX)")
plt.title("Average Contract Value by Region")
```

```
plt.xticks(rotation=45)
plt.show()
```



STEP 5: Data Cleaning & Preparation

```
df_ml = df.copy()

# Encode categorical variables
label_encoders = {}

categorical_cols = [
    'Procuring_Entity',
    'Category',
    'Item_Description',
    'Procurement_Method',
    'Region'
]

for col in categorical_cols:
    le = LabelEncoder()
    df_ml[col] = le.fit_transform(df_ml[col])
    label_encoders[col] = le

# Encode target variable
le_status = LabelEncoder()
df_ml['Status'] = le_status.fit_transform(df_ml['Status'])
```

STEP 6: Feature Selection

```
X = df_ml.drop(['Contract_ID', 'Status'], axis=1)
y = df_ml['Status']
```

STEP 7: Feature Scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

STEP 8: Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

STEP 9: Classical ML Model (Random Forest)

```
rf = RandomForestClassifier()

rf.fit(X_train, y_train)

rf_preds = rf.predict(X_test)

print("Random Forest Results:")
print(classification_report(y_test, rf_preds))
```

```
Random Forest Results:
              precision    recall  f1-score   support

           0       0.70      0.99      0.82       154
           1       0.50      0.04      0.07        54
           2       0.00      0.00      0.00        12

    accuracy                           0.70       220
   macro avg       0.40      0.34      0.30       220
weighted avg       0.62      0.70      0.59       220

/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

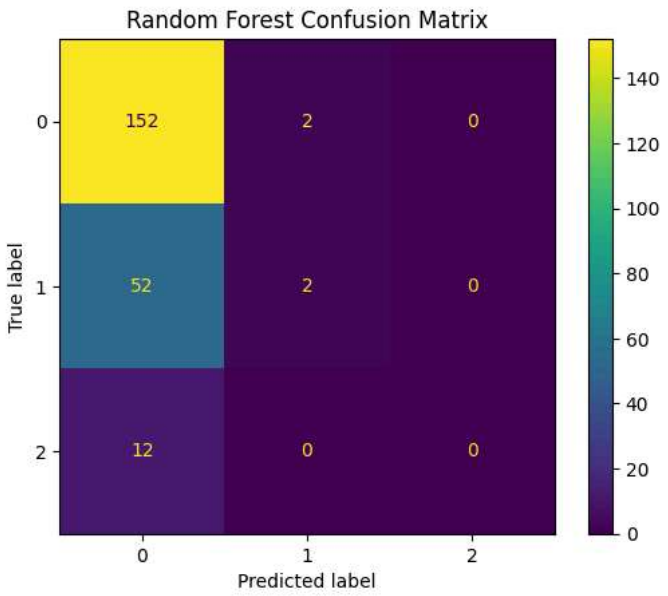
STEP 10: Neural Network Model

```
mlp = MLPClassifier(max_iter=500)

mlp.fit(X_train, y_train)

mlp_preds = mlp.predict(X_test)

print("Neural Network Results:")
print(classification_report(y_test, mlp_preds))
```

```
Neural Network Results:
              precision    recall  f1-score   support

           0       0.70      0.97      0.81       154
           1       0.17      0.02      0.03        54
           2       0.00      0.00      0.00        12

    accuracy                           0.68       220
   macro avg       0.29      0.33      0.28       220
weighted avg       0.53      0.68      0.58       220

/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:691: ConvergenceWarning: Stochastic Opt
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:1565: UndefinedMetricWarning: Precision is ill-define
  _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
```

STEP 11: Model Evaluation [Confusion Matrix]

```
from sklearn.metrics import ConfusionMatrixDisplay

ConfusionMatrixDisplay.from_estimator(rf, X_test, y_test)
plt.title("Random Forest Confusion Matrix")
plt.show()
```



WORKING DEMO:

```
# --- STEP: WORKING DEMO ---
import ipywidgets as widgets
from IPython.display import display, HTML

# 1. Setup UI Styling
style = {'description_width': 'initial'}
header = HTML("<h2>🚀 Procurement Status Prediction Demo</h2>")
display(header)

# 2. Dynamically Create Input Widgets for Features
input_widgets = {}

# We iterate through the columns used in training (X.columns)
for col in X.columns:
    if col in categorical_cols:
        # Create a dropdown for categorical features using the encoder classes
        input_widgets[col] = widgets.Dropdown(
            options=label_encoders[col].classes_,
            description=f'{col.replace("_", " ")}:',
            style=style,
            layout={'width': '400px'}
        )
    else:
        # Create a numeric input for numerical features
        input_widgets[col] = widgets.FloatText(
            value=float(df[col].median()), # Default to median value
            description=f'{col.replace("_", " ")}:',
            style=style,
            layout={'width': '400px'}
        )

# 3. Create Prediction Button and Output Area
predict_btn = widgets.Button(
    description="Predict Status",
    button_style='primary', # 'success', 'info', 'warning', 'danger' or ''
    tooltip='Click to predict the procurement status',
    icon='check'
)
output = widgets.Output()

# 4. Define the Prediction Logic
```

```
def perform_prediction(b):
    with output:
        output.clear_output()
        try:
            # Gather inputs into a dictionary
            data_dict = {}
            for col, widget in input_widgets.items():
                val = widget.value
                if col in categorical_cols:
                    # Transform the string input to the encoded integer
                    val = label_encoders[col].transform([val])[0]
                data_dict[col] = val

            # Convert to DataFrame to ensure correct column order
            input_df = pd.DataFrame([data_dict])[X.columns]

            # Apply the same scaling used during training
            input_scaled = scaler.transform(input_df)

            # Get prediction from the Random Forest model
            # (You can change 'rf' to 'mlp' if you prefer the Neural Network)
            pred_code = rf.predict(input_scaled)[0]
            prediction = le_status.inverse_transform([pred_code])[0]

            # Get probability scores
            probs = rf.predict_proba(input_scaled)[0]
            max_prob = max(probs) * 100

            # Display Result
            print("-" * 30)
            display(HTML(f"<h3>Result: <span style='color: #2e7d32;'>{prediction}</span></h3>"))
            print(f"Confidence: {max_prob:.2f}%")
            print("-" * 30)

        except Exception as e:
            display(HTML(f"<b style='color: red;'>Error:</b> {str(e)}"))

# Link button to function
predict_btn.on_click(perform_prediction)

# 5. Display the UI
ui_layout = widgets.VBox(list(input_widgets.values()) + [widgets.Label(""), predict_btn, output])
display(ui_layout)


import joblib
# Save the model and encoders
joblib.dump({'model': rf, 'scaler': scaler, 'encoders': label_encoders, 'target_encoder': le_status}, 'procurement_model.pkl')
```

🚀 Procurement Status Prediction Demo

Procuring Entity: Kampala Capital City Authority (KCCA)

Category: Consultancy Services

Item Description: Auditing Services

Procurement Method: Direct Procurement

Contract Value UGX: 162831174.5

Number of Bids: 5

Region: Central

Predict Status

```
['procurement_model.pkl']
```